A Vanishing Proportion of Metrizable Path Systems on 2-Connected Graphs

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Abstract

For a finite connected graph G, let $\mathfrak{P}(G)$ be the set of consistent path systems in G, and let $\mathfrak{M}(G) \subseteq \mathfrak{P}(G)$ be the systems induced by positive edge weights, with ties among shortest paths allowed. Cizma and Linial asked whether there is a sequence of 2-connected n-vertex graphs G_n for which $|\mathfrak{M}(G_n)|/|\mathfrak{P}(G_n)| \to 0$. We answer this question affirmatively. Let Γ_m consist of one s–t path of length three and m internally disjoint s–t paths of length two. Then Γ_m is 2-connected, $|V(\Gamma_m)| = m + 4$, and

$$\frac{|\mathfrak{M}(\Gamma_m)|}{|\mathfrak{P}(\Gamma_m)|} = O\Big(m^3 \mathrm{e}^{-m/24}\Big).$$

The proof is combinatorial and probabilistic. We construct $3^m 2^{\binom{m}{2}}$ distinct consistent path systems, encode every full-support system by three vertex types and a two-colouring of pairs, and exhibit a four-inequality obstruction that excludes almost all such encodings from being metrizable. A separate support-structure argument shows that systems omitting at least one edge are asymptotically negligible. The proof does not depend on computer enumeration; a bounded independent verification script is supplied as supplementary material.

Keywords. graph metrizability; consistent path system; shortest path; asymptotic enumeration; probabilistic method; 2-connected graph.
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1 Introduction

A path system in a finite connected graph G chooses one simple path $P_{u,v}$ between every unordered pair of vertices. It is *consistent* if every subpath of a chosen path is itself the chosen path between its endpoints. A positive edge weighting *induces* the system if every chosen path is a shortest path; ties are allowed. Cizma and Linial initiated the systematic study of graphs for which every consistent path system is induced in this sense [2]. We call an individual system with such an inducing weighting *metrizable*; this is the same notion called a *metric path system* in [5].

For a fixed graph G, write

$$\mathfrak{P}(G) = \{\text{consistent path systems in } G\}, \qquad \mathfrak{M}(G) = \{\mathcal{P} \in \mathfrak{P}(G) : \mathcal{P} \text{ is metrizable}\}.$$

The inclusion $\mathfrak{M}(G) \subseteq \mathfrak{P}(G)$ refines the binary question of whether G itself is metrizable. In Open Problem 11.4 of *Geodesic Geometry on Graphs*, Cizma and Linial asked whether there are 2-connected n-vertex graphs for which the proportion of metrizable systems tends to zero [2, Open Problem 11.4].

Subsequent work developed a structural theory of metrizable graphs [1], constructed irreducible non-metrizable systems [3], and introduced quantitative metric approximation [4].

Cizma and Linial also determined the logarithmic order of the global numbers of consistent and metrizable systems on an n-element vertex set and conjectured that their global ratio tends to zero [5, Sec. 7]. That global question is logically distinct from the fixed-graph existence problem considered here.

For $m \geq 1$, let Γ_m be the graph with vertex set

$$V(\Gamma_m) = \{s, t, a, b, x_1, \ldots, x_m\}$$

and edge set

$$E(\Gamma_m) = \{sa, ab, bt\} \cup \{sx_i, tx_i : 1 \leq i \leq m\}. \tag{1.1}$$

Thus $s - a - b - t$ is one s–t arm of length three, while every $s - x_i - t$ is an arm of length two; see Figure 1.

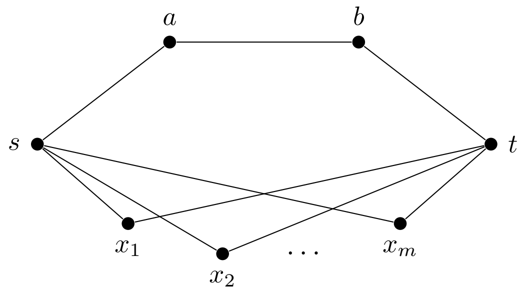

Figure 1: The graph Γ_m, with $|V(\Gamma_m)| = m + 4$ and $|E(\Gamma_m)| = 2m + 3$.

Our main result is deliberately stated before the more detailed numerical bound.

Theorem 1.1. *As $m \to \infty$,*

$$\frac{|\mathfrak{M}(\Gamma_m)|}{|\mathfrak{P}(\Gamma_m)|} = O\Big(m^3 \mathrm{e}^{-m/24}\Big)\,.$$

In particular, this ratio tends to zero.

Corollary 1.2 (Open Problem 11.4)**.** *There exists a sequence $(G_n)_{n \geq 5}$ of 2-connected n-vertex graphs such that*

$$\lim_{n \to \infty} \frac{|\mathfrak{M}(G_n)|}{|\mathfrak{P}(G_n)|} = 0.$$

One may take $G_n = \Gamma_{n-4}$.

The proof has three ingredients. First, we exhibit $3^m 2^{\binom{m}{2}}$ distinct consistent systems, giving a lower bound for the denominator. Second, every full-support system has a compact encoding by three vertex types and a two-colouring of the pairs $\{x_i, x_j\}$. A local four-inequality cancellation certifies non-metrizability for a simple three-vertex pattern, and a concentration argument shows that almost every encoding contains that pattern. Third, the support graph of a system that omits an edge has only one possible cyclic block: either a smaller member of the same family or a complete bipartite graph. This makes all proper supports jointly negligible. The constants are not optimized.

2 Support and block reductions

All graphs are finite, simple, and connected, and all paths are simple. A path system is indexed by unordered vertex pairs, so a path and its reverse are identified. If $w : E(G) \to (0, \infty)$ and Q is a path, write $w(Q) = \sum_{e \in E(Q)} w(e)$.

Definition 2.1. A path system $\mathcal{P} = \{P_{u,v} : \{u,v\} \in \binom{V(G)}{2}\}$ is *consistent* if, whenever $x, y \in V(P_{u,v})$, the x–y subpath of $P_{u,v}$ equals $P_{x,y}$. It is *neighborly* if $P_{u,v} = uv$ for every edge $uv \in E(G)$. It is *metrizable* if some positive edge weighting makes every $P_{u,v}$ a shortest u–v path.

Let $\mathfrak{N}(G)$ denote the set of neighborly consistent path systems in G. For a path system $\mathcal{P}$ in G, define its *support graph* by

$$\operatorname{supp}(\mathcal{P}) = \left(V(G), \bigcup_{\{u,v\} \in \binom{V(G)}{2}} E(P_{u,v}) \right). \tag{2.1}$$

Lemma 2.2 (Support reduction)**.** *Let $\mathcal{P}$ be a consistent path system in G and put $H = \operatorname{supp}(\mathcal{P})$. Then:*

(i) *H is a connected spanning subgraph of G;*

(ii) *$\mathcal{P}$, viewed as a path system in H, is neighborly;*

(iii) *$\mathcal{P}$ is metrizable in G if and only if it is metrizable in H.*

Proof. The union of all chosen paths contains every vertex and connects every pair, so it is a connected spanning subgraph. If $xy \in E(H)$, then the edge xy occurs as a subpath of some chosen path. Consistency therefore forces $P_{x,y} = xy$, proving neighborliness.

If a weighting of G induces $\mathcal{P}$, its restriction to H does so. Conversely, suppose that w induces $\mathcal{P}$ in H. Assign every edge of $E(G) \setminus E(H)$ a common weight greater than $\sum_{e \in E(H)} w(e)$. Every chosen path lies in H and has weight at most this sum, whereas every competing path using an edge outside H has larger weight. Hence the extended weighting induces $\mathcal{P}$ in G. ☐

Corollary 2.3. *For every connected spanning subgraph H of G, the path systems in G whose support is exactly H are, by restriction, in bijection with $\mathfrak{N}(H)$.*

Proof. A system with support H is neighborly on H by Lemma 2.2. Conversely, every neighborly system on H, viewed as a system in G, uses every edge of H and no edge outside H, so its support is exactly H. ☐

We use the standard convention that a block of a connected graph is either a maximal 2-connected subgraph or a bridge together with its endpoints.

Lemma 2.4 (Block product)**.** *Let $B_1, \dots, B_q$ be the blocks of a connected graph H. Restriction to the blocks is a bijection*

$$\mathfrak{P}(H) \longrightarrow \prod_{j=1}^{q} \mathfrak{P}(B_j). \tag{2.2}$$

The same statement holds with $\mathfrak{P}$ replaced by $\mathfrak{N}$. In particular, bridge blocks contribute a factor one.

Proof. Let T be the block-cut tree of H, whose nodes are the blocks and the cut vertices, with incidence recording containment. If two vertices u, v lie in the same block B, then every simple u–v path of H is contained in B. Indeed, an excursion leaving B through a cut vertex and returning through the same cut vertex would repeat that vertex, while an excursion returning through a different cut vertex would give, in T, a path from B back to B of positive length. Both alternatives are impossible. Thus every global path system restricts to a path system on each block, and consistency is inherited.

Conversely, choose a consistent path system on every block. Associate with a vertex u the cut-vertex node u of T when u is a cut vertex, and otherwise the unique block node containing

u. The unique route in T between the nodes associated with u and v determines an ordered sequence of blocks and intervening cut vertices. In the first and last block use u or v as the corresponding endpoint, and in every intermediate block use the two adjacent cut vertices. Concatenating the selected within-block paths for these endpoint pairs gives a simple u–v path: consecutive pieces meet at exactly their common cut vertex, and nonconsecutive pieces lie in distinct nodes of the tree route.

If x, y occur on this concatenated path, its x–y subpath traverses a consecutive subroute of the same block-cut route. Within its first and last blocks it is a selected subpath by local consistency, and every intermediate block piece is unchanged. Hence it is exactly the path constructed from the local systems for x, y. The global system is therefore consistent. The two constructions are inverse, and neighborliness is preserved blockwise. A bridge block has only its one-edge path system. □

Lemma 2.5. *Every chosen path in a neighborly consistent path system is an induced path.*

Proof. If a chosen path contained a chord xy, then consistency would identify $P_{x,y}$ with the nontrivial x–y subpath of the chosen path, while neighborliness would give $P_{x,y} = xy$, a contradiction. □

3 A large explicit family

Lemma 3.1. *The graph Γ_m is 2-connected for every $m \geq 1$.*

Proof. The graph is connected. After deleting s or t, all remaining vertices are connected through the other branch vertex and the path $a - b$; after deleting a or b, the vertices remain connected through any length-two arm $s - x_i - t$; and after deleting any x_i, the core path $s - a - b - t$ remains. Thus Γ_m has no cut vertex. □

We now construct many consistent path systems. Every edge is prescribed as the path between its endpoints, and we fix

$$P_{s,t} = sabt, \qquad P_{s,b} = sab, \qquad P_{a,t} = abt. \tag{3.1}$$

For each $i \in [m]$, independently choose a type $\tau_i \in \{\mathrm{A}, \mathrm{B}, \mathrm{C}\}$ and prescribe

$$\begin{array}{c|cc} \tau_i & P_{a,x_i} & P_{b,x_i} \\ \hline \mathrm{A} & asx_i & basx_i \\ \mathrm{B} & abtx_i & btx_i \\ \mathrm{C} & asx_i & btx_i \end{array} \tag{3.2}$$

For every $1 \leq i < j \leq m$, independently choose

$$P_{x_i,x_j} = x_i s x_j \quad \text{or} \quad P_{x_i,x_j} = x_i t x_j. \tag{3.3}$$

The following table records all unordered pair types and the subpaths that must be checked.

Proposition 3.2. *Every choice in* (3.1)–(3.3) *produces a consistent path system, and distinct choices produce distinct systems. Consequently,*

$$|\mathfrak{P}(\Gamma_m)| \geq L_m := 3^m 2^{\binom{m}{2}}. \tag{3.4}$$

Proof. The prescriptions cover every unordered vertex pair. Indeed, they cover the $2m+3$ edges, the three non-edge core pairs $(s,t), (s,b), (a,t)$, the $2m$ pairs $\{a, x_i\}, \{b, x_i\}$, and the $\binom{m}{2}$ pairs $\{x_i, x_j\}$; the total is

$$(2m+3) + 3 + 2m + \binom{m}{2} = \binom{m+4}{2}.$$

Table 1: The prescriptions in the explicit family.

Endpoint pair	Prescribed path	Nontrivial proper subpaths to verify
an edge of Γ_m	the edge itself	none
s, t	$sabt$	sab and abt
s, b or a, t	sab or abt	edges only
a, x_i and b, x_i	one of the three rows of (3.2)	for type A, the path $basx_i$ has subpaths bas and asx_i; for type B, use the symmetric paths abt and btx_i; type C uses two-edge paths
x_i, x_j	x_isx_j or x_itx_j	edges only

It remains to check subpath closure. The path $sabt$ has the prescribed subpaths sab and abt. For a type-A vertex, the only path of length greater than two is $basx_i$; its non-edge proper subpaths are bas, the reverse of $P_{s,b}$, and $asx_i = P_{a,x_i}$. The type-B verification is symmetric. Every type-C path and every path in (3.3) has length at most two. All remaining proper subpaths are edges. Thus every subpath of every prescribed path is prescribed correctly, so the system is consistent.

Changing τ_i changes at least one of P_{a,x_i}, P_{b,x_i}, and changing a pair choice changes P_{x_i,x_j}. Hence the encoding is injective and the number of systems is exactly $3^m 2^{\binom{m}{2}}$ within this family. $\square$

4 Full supports and a local obstruction

By Lemma 2.2, a system on Γ_m has full support if and only if it is neighborly. We first make the induced-path catalogue explicit.

Lemma 4.1 (Induced-path catalogue)**.** *The induced paths in Γ_m between nonadjacent vertices are precisely the following:*

$$\begin{aligned} s\text{–}t: &\quad sabt, \quad sx_it \ (i \in [m]); \\ s\text{–}b: &\quad sab, \quad sx_itb \ (i \in [m]); \\ a\text{–}t: &\quad abt, \quad asx_it \ (i \in [m]); \\ a\text{–}x_i: &\quad asx_i, \quad abtx_i; \\ b\text{–}x_i: &\quad btx_i, \quad basx_i; \\ x_i\text{–}x_j: &\quad x_isx_j, \quad x_itx_j \quad (i \neq j). \end{aligned}$$

Proof. An s–t path cannot change arms because distinct arms meet only at s, t; hence it is one of the displayed arms. The descriptions for s–b and a–t follow in the same way by appending or deleting the last core edge.

Consider an a–x_i path. If it leaves a through s and does not use sx_i immediately, then it must reach t along another arm before using tx_i; the edge sx_i is then a chord. If it leaves a through b, the only chordless possibility is $abtx_i$. Thus the two displayed paths are exactly the induced ones. The b–x_i case is symmetric. Finally, an x_i–x_j path not equal to one of the two displayed length-two paths must travel from one branch vertex to the other before reaching its second endpoint; then either x_it or sx_j (or symmetrically x_is or tx_j) is a chord. This proves the list. $\square$

Lemma 4.2 (Encoding lemma)**.** *The number of full-support consistent path systems on Γ_m is at most*

$$(m+1)^3 3^m 2^{\binom{m}{2}}. \tag{4.1}$$

More precisely, every such system determines:

(i) *one of at most $m+1$ choices for each of $P_{s,t}$, $P_{s,b}$, and $P_{a,t}$;*

(ii) *a type $\tau_i \in \{\mathrm{A}, \mathrm{B}, \mathrm{C}\}$ for every $i \in [m]$, as in* (3.2)*;*

(iii) *a colour $c_{ij} \in \{s, t\}$ for every $i < j$, according as P_{x_i,x_j} passes through s or t.*

For each fixed triple $(P_{s,t}, P_{s,b}, P_{a,t})$, the map from a path system to its type-colour data is injective.

Proof. A full-support system is neighborly, so every chosen path is induced by Lemma 2.5. The catalogue in Lemma 4.1 gives at most $m+1$ possibilities for each of the three core paths. For each i,

$$P_{a,x_i} \in \{asx_i, abtx_i\}, \qquad P_{b,x_i} \in \{btx_i, basx_i\}.$$

The combination $P_{a,x_i} = abtx_i$ and $P_{b,x_i} = basx_i$ is impossible: the a–x_i subpath of $basx_i$ is asx_i, so consistency would force $P_{a,x_i} = asx_i$. The remaining three combinations are exactly the types in (3.2). The only induced x_i–x_j paths are the two length-two paths through s and t.

Once the three core paths are fixed, these types and colours specify every remaining non-edge pair, while the edge pairs are fixed by neighborliness. Thus the map is injective. Multiplication gives (4.1). □

Let

$$\Omega_m = \{\mathrm{A}, \mathrm{B}, \mathrm{C}\}^{[m]} \times \{s, t\}^{\binom{[m]}{2}}. \tag{4.2}$$

An element of Ω_m consists of the type and pair-colour data from Lemma 4.2, and $|\Omega_m| = L_m$. We use the symmetric notation $c_{ij} = c_{ji}$ when $i > j$. Call an encoding *bad* if there are distinct indices i, j, k such that

$$\tau_i = \mathrm{A}, \qquad \tau_j = \mathrm{B}, \qquad \tau_k = \mathrm{C}, \qquad c_{ik} = t, \qquad c_{jk} = s. \tag{4.3}$$

The type-C requirement on the centre k is not needed in the algebraic contradiction below. Its purpose is probabilistic: after the type partition is fixed, different type-C centres involve disjoint sets of pair-colour variables.

Lemma 4.3 (Four-inequality obstruction)**.** *No metrizable full-support path system on Γ_m has a bad encoding.*

Proof. Let $x = x_i$, $y = x_j$, and $z = x_k$ witness (4.3), and write $w_{uv} = w(uv) = w(vu)$. The selected paths xtz and ysz, compared with the routes through the opposite branch vertex, give

$$\begin{aligned} w_{tx} + w_{tz} &\le w_{sx} + w_{sz}, \\ w_{sy} + w_{sz} &\le w_{ty} + w_{tz}. \end{aligned} \tag{4.4}$$

Since y has type B, the selected path $abty$ is no longer than asy; since x has type A, the selected path $basx$ is no longer than btx. Hence

$$\begin{aligned} w_{ab} + w_{bt} + w_{ty} &\le w_{as} + w_{sy}, \\ w_{ab} + w_{as} + w_{sx} &\le w_{bt} + w_{tx}. \end{aligned} \tag{4.5}$$

Adding the four inequalities and cancelling all terms except the two copies of w_{ab} yields $2w_{ab} \le 0$, contrary to the positivity of the edge weights. □

We now count encodings that avoid (4.3).

Proposition 4.4. *For $m \ge 12$, the number of non-bad encodings in Ω_m is at most*

$$\varepsilon_m L_m, \qquad \varepsilon_m := 3\mathrm{e}^{-m/24} + 2^{-m^2/36 + m/6}. \tag{4.6}$$

Proof. Choose an encoding uniformly at random. Let $I_{\mathrm{A}}, I_{\mathrm{B}}, I_{\mathrm{C}}$ be the sets of indices of types $\mathrm{A}, \mathrm{B}, \mathrm{C}$, and put $p = |I_{\mathrm{A}}|$, $q = |I_{\mathrm{B}}|$, and $r = |I_{\mathrm{C}}|$. Each of p, q, r has marginal distribution $\mathrm{Bin}(m, 1/3)$. The multiplicative Chernoff bound with mean $m/3$ and relative deviation $1/2$ gives $\Pr(p < m/6) \le \mathrm{e}^{-m/24}$, and the same estimate holds for q and r. Therefore

$$\Pr\left(\min\{p, q, r\} < \frac{m}{6}\right) \le 3\mathrm{e}^{-m/24}. \tag{4.7}$$

Condition on a fixed type partition with $p, q, r \ge m/6$. For a fixed $k \in I_{\mathrm{C}}$, let E_k be the event that no bad triple is centred at k. Then

$$E_k = \{c_{ik} = s \text{ for all } i \in I_{\mathrm{A}}\} \;\cup\; \{c_{jk} = t \text{ for all } j \in I_{\mathrm{B}}\}.$$

The two events in this union depend on disjoint colour variables, so

$$\Pr(E_k \mid I_{\mathrm{A}}, I_{\mathrm{B}}, I_{\mathrm{C}}) = 2^{-p} + 2^{-q} - 2^{-(p+q)} \le 2^{-p} + 2^{-q} \le 2^{1-m/6}. \tag{4.8}$$

If $k, k' \in I_{\mathrm{C}}$ are distinct, then E_k and $E_{k'}$ depend on disjoint sets of unordered-pair colours: one set consists of pairs joining k to $I_{\mathrm{A}} \cup I_{\mathrm{B}}$, and the other consists of pairs joining k' to the same set. Thus the events $(E_k)_{k \in I_{\mathrm{C}}}$ are conditionally independent. Since $m \ge 12$ and $r \ge m/6$,

$$\begin{aligned}\Pr(\text{non-bad} \mid I_{\mathrm{A}}, I_{\mathrm{B}}, I_{\mathrm{C}}) &= \prod_{k \in I_{\mathrm{C}}} \Pr(E_k \mid I_{\mathrm{A}}, I_{\mathrm{B}}, I_{\mathrm{C}}) \\ &\le \left(2^{1-m/6}\right)^r \le 2^{(m/6)(1-m/6)} = 2^{-m^2/36 + m/6}. \end{aligned} \tag{4.9}$$

Combining (4.7) and (4.9) proves (4.6). □

Corollary 4.5. *For $m \ge 12$, the number M_m^{full} of metrizable full-support systems on Γ_m satisfies*

$$M_m^{\text{full}} \le (m+1)^3 \varepsilon_m L_m. \tag{4.10}$$

Proof. Fix one of the at most $(m+1)^3$ triples of core paths in Lemma 4.2. The type-colour encoding is injective for this fixed triple, and a metrizable system must map to a non-bad encoding by Lemma 4.3. Apply Proposition 4.4 and then sum over the core triples. □

5 Systems with proper support

We now classify the connected spanning supports of Γ_m. For such a support H, define

$$R(H) = \{i \in [m] : sx_i, tx_i \in E(H)\}, \qquad r = |R(H)|. \tag{5.1}$$

Lemma 5.1 (Structure of connected supports)**.** *Let H be a connected spanning subgraph of Γ_m.*

(i) *If $i \notin R(H)$, then x_i is a leaf of H.*

(ii) *If $sa, ab, bt \in E(H)$ and $r \ge 1$, then the unique cyclic block of H is the copy of Γ_r induced by the core arm and the full length-two arms indexed by $R(H)$. If $r = 0$, H is a tree.*

(iii) *If at least one of sa, ab, bt is absent and $r \ge 2$, then the unique cyclic block of H is the copy of $K_{2,r}$ on $\{s, t\} \cup \{x_i : i \in R(H)\}$. If $r \le 1$, H is a tree.*

All edges outside the stated cyclic block are bridges.

Proof. Each x_i has exactly two possible incident edges in Γ_m. Since H is connected and spanning, an index outside $R(H)$ retains exactly one of these edges, so x_i is a leaf.

The graph $\Gamma_m - \{s,t\}$ consists of the path $a - b$ and the isolated vertices x_i. Consequently every cycle of Γ_m, and hence every cycle of H, is the union of two distinct complete s–t arms. The core arm $s - a - b - t$ is available exactly when all three core edges are present; the short arm $s - x_i - t$ is available exactly when $i \in R(H)$.

If the core arm is available and $r \geq 1$, the union of that arm with all available short arms is Γ_r, which is 2-connected by Lemma 3.1; every cycle lies in this subgraph. If the core arm is unavailable, cycles can use only pairs of available short arms. For $r \geq 2$, their union is $K_{2,r}$, which is 2-connected; for $r \leq 1$ no cycle exists. An edge lying on no cycle in a connected graph is a bridge, which proves the final assertion. □

For counting, let Γ_0 denote the path $s - a - b - t$ and set

$$A_r = (r+1)^3 3^r 2^{\binom{r}{2}} \qquad (r \geq 0). \tag{5.2}$$

By Lemma 4.2, $|\mathfrak{N}(\Gamma_r)| \leq A_r$ for $r \geq 1$, and the same bound is immediate for $r = 0$. For the complete bipartite block, put

$$B_r = \max\{1, r\} 2^{\binom{r}{2}} \qquad (r \geq 0). \tag{5.3}$$

Then $|\mathfrak{N}(K_{2,r})| \leq B_r$ for $r \geq 2$, while $B_0 = B_1 = 1$ represents the absence of a cyclic block. To see the bound for $r \geq 2$, choose the selected s–t path through one of the r degree-two vertices and, for every pair of such vertices, choose the route through s or through t. These data exhaust all non-edge pairs and therefore give an upper bound; consistency may eliminate some choices. In all cases $B_r \leq A_r$.

Proposition 5.2. *Let N_m^{proper} be the number of consistent path systems on Γ_m whose support is a proper subgraph. Then*

$$N_m^{\text{proper}} \leq 7m 2^{\binom{m}{2}} + 8 \sum_{r=0}^{m-1} \binom{m}{r} 2^{m-r} (r+1)^3 3^r 2^{\binom{r}{2}}. \tag{5.4}$$

Consequently,

$$\frac{N_m^{\text{proper}}}{L_m} \leq \frac{7m}{3^m} + 8(m+1)^3 \left[\left(1 + \frac{2}{3} 2^{-(m-1)/2} \right)^m - 1 \right] = o(1). \tag{5.5}$$

Proof. Fix $0 \leq r < m$. There are $\binom{m}{r}$ choices for $R(H)$. Every remaining x_i is a leaf and has two choices for its unique incident support edge, giving at most 2^{m-r} possibilities. There are at most 8 subsets of the three core edges. Some resulting edge sets are disconnected; retaining them only enlarges the upper bound. By Corollary 2.3, the systems with support H are counted by $\mathfrak{N}(H)$. The block product Lemma 2.4, the structural classification Lemma 5.1, and the bounds $B_r \leq A_r$ show that each such support contributes at most A_r systems. This gives the sum in (5.4).

If $r = m$ and the support is proper, then every edge incident with an x_i is present and at least one core edge is absent. There are at most seven core edge subsets, and Lemma 5.1 leaves only the block $K_{2,m}$ together with bridge attachments. Hence this case contributes at most $7B_m \leq 7m2^{\binom{m}{2}}$ systems, the first term of (5.4).

Divide by $L_m = 3^m 2^{\binom{m}{2}}$ and put $k = m - r$. Since

$$\binom{m}{2} - \binom{m-k}{2} = \frac{k(2m-k-1)}{2} \geq \frac{k(m-1)}{2}, \tag{5.6}$$

the normalized sum is at most

$$8(m+1)^3 \sum_{k=1}^{m} \binom{m}{k} \left(\frac{2}{3}\, 2^{-(m-1)/2}\right)^k.$$

The binomial theorem gives the bracketed expression in (5.5). Put $\alpha_m = (2/3)2^{-(m-1)/2}$. Since $m\alpha_m \to 0$, for all sufficiently large m we have $0 \le m\alpha_m \le 1$; hence

$$(1+\alpha_m)^m - 1 \le \mathrm{e}^{m\alpha_m} - 1 \le 2m\alpha_m.$$

Thus the second term in (5.5) is $O(m^4 2^{-m/2})$, while the first is $O(m3^{-m})$. Both tend to zero. $\square$

6 Quantitative bound and completion of the proof

Proposition 6.1 (Explicit quantitative bound)**.** *For every $m \ge 12$,*

$$\frac{|\mathfrak{M}(\Gamma_m)|}{|\mathfrak{P}(\Gamma_m)|} \le (m+1)^3 \left(3\mathrm{e}^{-m/24} + 2^{-m^2/36+m/6}\right) + \frac{7m}{3^m} + 8(m+1)^3 \left[\left(1 + \frac{2}{3}\, 2^{-(m-1)/2}\right)^m - 1\right]. \tag{6.1}$$

Proof. Every metrizable system has either full support or proper support. Hence, by Corollary 4.5 and Proposition 5.2,

$$|\mathfrak{M}(\Gamma_m)| \le M_m^{\mathrm{full}} + N_m^{\mathrm{proper}}.$$

On the other hand, Proposition 3.2 gives $|\mathfrak{P}(\Gamma_m)| \ge L_m$. Dividing by L_m and substituting (4.6), (4.10), and (5.5) yields (6.1). $\square$

Proof of Theorem 1.1. The first summand in (6.1) is $O(m^3\mathrm{e}^{-m/24})$, and the term containing $2^{-m^2/36+m/6}$ is superexponentially small. By Proposition 5.2, the remaining contribution is

$$O(m^4 2^{-m/2}) + O(m3^{-m}) = O(m^3\mathrm{e}^{-m/24}).$$

This proves the asserted estimate and its vanishing limit. $\square$

Proof of Corollary 1.2. For $n \ge 5$, take $G_n = \Gamma_{n-4}$. By Lemma 3.1, G_n is 2-connected and has n vertices. The required limit is exactly the limit in Theorem 1.1 with $m = n - 4$. $\square$

Corollary 6.2. *The proportion of strictly metrizable systems in $\mathfrak{P}(\Gamma_m)$ also tends to zero.*

Proof. Every strictly metrizable system is metrizable, so its proportion is bounded above by the ratio in Theorem 1.1. $\square$

7 Concluding remarks

The proof separates two mechanisms. On full support, the obstruction in Lemma 4.3 is local and appears with exponentially high probability in the type-colour encoding. On proper supports, missing edges turn most x_i into leaves, and the only cyclic block is a smaller member of the same family or a complete bipartite graph. This self-similar support structure makes the total proper-support contribution summable.

The result proves the existence assertion in Open Problem 11.4. It does not determine the true exponential rate of $|\mathfrak{M}(\Gamma_m)|/|\mathfrak{P}(\Gamma_m)|$, nor does it settle the stronger claims that the same phenomenon holds for most n-vertex graphs or that the global ratio $|\mathfrak{M}_n|/|\mathfrak{P}_n|$ tends to zero. Natural next questions are to sharpen the rate for Γ_m, find denser 2-connected examples, and determine the behaviour for random graphs.

A Independent bounded verification

The proof above is independent of computation. The supplementary archive contains a pure-Python script, using only the standard library, that checks four bounded components where transcription errors would be particularly easy:

(i) all explicit type-colour systems are consistent and distinct for $1 \leq m \leq 4$;

(ii) the induced-path catalogue of Lemma 4.1 is exhaustive for $1 \leq m \leq 4$;

(iii) the coefficient sum of the four inequalities in Lemma 4.3 is exactly $2w_{ab}$;

(iv) every connected spanning support has the cyclic-edge structure of Lemma 5.1 for $1 \leq m \leq 6$.

It also checks the exact single-centre probability $2^{-p} + 2^{-q} - 2^{-(p+q)}$ for $0 \leq p, q \leq 5$. The recorded output is

m	1	2	3	4
explicit systems checked	3	18	216	5184

and the numbers of connected spanning supports checked for $m = 1, \ldots, 6$ are

$$6,\ 24,\ 84,\ 276,\ 876,\ 2724.$$

The command `verification/run_verification.sh` reproduces the log. These computations are audits only and are not invoked in any proof.

Declarations

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Data and code availability. No research data were generated. The supplementary submission archive contains the pure-Python verification script, its execution log, and a SHA-256 manifest. The mathematical proof does not depend on these checks.

Competing interests. The author declares no competing financial or non-financial interests.

Use of generative AI. We used ChatGPT to brainstorm the structure, anticipate potential objections, and refine the text. We are fully responsible for all claims, citations, calculations, and the final wording.

Author contributions. Guangfu Wang: Conceptualization, methodology, formal analysis, investigation, writing–original draft, writing–review and editing, and funding acquisition.

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